

Tabular Machine-Learning Vetting of NASA Kepler Objects of Interest: A Reproducible Five-Classifier Comparison

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Abstract—We present a reproducible machine-learning pipeline that classifies Kepler Objects of Interest (KOIs) from the NASA Exoplanet Archive Cumulative Table as either genuine planet *candidates* or *false positives* using only the photometric and stellar features available *before* the official vetting decision is made. Five classifier families with deliberately different inductive biases — Logistic Regression, k -Nearest Neighbours, Random Forest, XGBoost and a Multi-Layer Perceptron — are trained on a stratified 72/8/20 split of 9 564 KOIs after explicit removal of label-leaking columns. SMOTE is conditionally applied based on the empirical minority/majority ratio. Hyper-parameters of the two strongest non-linear models are tuned with 5-fold cross-validated GridSearchCV on macro F1. The tuned XGBoost classifier achieves the highest test-set performance ($F_1 = 0.860$, ROC-AUC = 0.936, MCC = 0.721), followed by Random Forest, MLP, Logistic Regression and KNN. Threshold tuning via Youden’s J statistic increases recall to 0.899 at a small precision cost — the right operating point when missing a real planet is more costly than a wasted follow-up. The full code, the executed notebooks, the saved best pipeline and the 14 figures referenced here are released under Apache 2.0 and reproduce deterministically with `RANDOM_SEED = 42`. The work continues a Global Finalist entry of the NASA Space Apps Challenge 2025.

Index Terms—Exoplanet detection, Kepler, KOI, binary classification, XGBoost, Random Forest, MLP, SMOTE, GridSearchCV, MCC, ROC, reproducible machine learning.

I. INTRODUCTION

The Kepler space telescope monitored the brightness of more than 150 000 stars over its primary mission and produced almost 10 000 *Kepler Objects of Interest* (KOIs): periodic dimming events that *may* correspond to a planet transiting in front of its host star, but that are equally consistent with eclipsing binaries, stellar variability or instrumental artefacts [1], [15]. Each KOI is curated in the NASA Exoplanet Archive Cumulative Table [2] together with photometric transit parameters, stellar properties of the host, and a final vetting disposition produced by a partly manual NASA pipeline.

Automating the triage step — deciding whether a KOI is worth promoting to a follow-up observation queue — has direct scientific impact. A reliable classifier reduces the human-vetting load and accelerates the subsequent confirmation campaigns [3], [5]. The objective of this paper is therefore to produce, evaluate and *reproduce* a strong tabular baseline for that task while explicitly avoiding the data-leakage trap that has inflated metrics in earlier studies.

The work also continues, with full course-grade methodology, the *Exoplanet Hunter AI* prototype that the ECI Centauri team submitted to the NASA Space Apps Challenge 2025 and that earned a Global Finalist position. That hackathon system shipped a working web application and a light-curve CNN in 48 hours; here we focus on the tabular vetting question and on the rigour required by the *MLEA_M* — *Machine Learning* course of the Master in Data Science at Escuela Colombiana de Ingeniería Julio Garavito (ECI).

Contributions: The paper makes four concrete contributions: (i) an explicit list of nine label-leaking columns that must be removed from the KOI cumulative table; (ii) a head-to-head comparison of five algorithm families with different inductive biases; (iii) hyper-parameter tuning and threshold optimisation on the two strongest non-linear families with a single primary metric (F_1); and (iv) a fully reproducible open-source release including 63 unit tests, the executed notebooks, all 14 figures used here, and a serialised joblib pipeline.

II. RELATED WORK

A. Tabular vs. light-curve classification

Two streams of work address KOI classification: *tabular* models that consume engineered features from the Kepler pipeline (this work, the *Robovetter* family [4]) and *light-curve* models that consume raw flux time-series with convolutional or transformer architectures [3], [5]. The two are

TABLE I: Target distribution before any preprocessing.

Class	Count	Fraction
FALSE POSITIVE	4 847	50.7%
CANDIDATE	4 717	49.3%
Total	9 564	100.0%

complementary; tabular models are cheap and explainable, light-curve models capture morphology that the table cannot.

B. Tree-based methods on tabular data

Gradient-boosted decision trees have dominated tabular benchmarks for two decades [7], [8], while Random Forest [6] provides a strong bagging baseline. Recent foundation-model approaches such as TabPFN [9] reach or exceed gradient-boosted performance on small datasets ($\leq 10\,000$ samples) without explicit hyper-parameter search; we discuss them as future work in Section X but keep XGBoost as the headline tuned model because it is the de-facto industry choice and is what the course covered.

C. Class imbalance

The KOI cumulative table is moderately imbalanced ($\approx 49\%/51\%$). When the minority/majority ratio falls below a tunable threshold we apply SMOTE [11] on the training set only — a guard against the inflation of cross-validated metrics that naive over-sampling produces.

III. DATASET AND LEAKAGE REMOVAL

The raw input is the NASA Exoplanet Archive Cumulative Table [2] retrieved on 2024-10-05 (9 564 rows $\times$ 49 columns, public-domain license).

A. Target

The label is the column `koi_pdisposition`, a binary indicator whose distribution is summarised in Table I. CANDIDATE is encoded as positive class 1 and FALSE POSITIVE as negative class 0.

B. Leakage removal

Nine columns either encode the label directly or are produced *after* the vetting decision and would therefore inflate test-set metrics without contributing to genuine generalisation. We drop them before any further processing:

- `koi_disposition` — final NASA disposition (label).
- `koi_score` — NASA’s own probability score.
- `koi_fpflag_nt`, `koi_fpflag_ss`, `koi_fpflag_co`, `koi_fpflag_ec` — false positive flags raised during the same vetting run.
- Identifier columns: `kepid`, `kepoi_name`, `kepler_name`, `koi_comment`.

The removal leaves thirteen physically meaningful numeric features spanning transit photometry (`koi_period`, `koi_time0bk`, `koi_impact`, `koi_duration`, `koi_depth`, `koi_prad`, `koi_teq`, `koi_insol`, `koi_model_snr`) and stellar properties (`koi_steff`, `koi_slogg`, `koi_srad`, `koi_kepmag`).

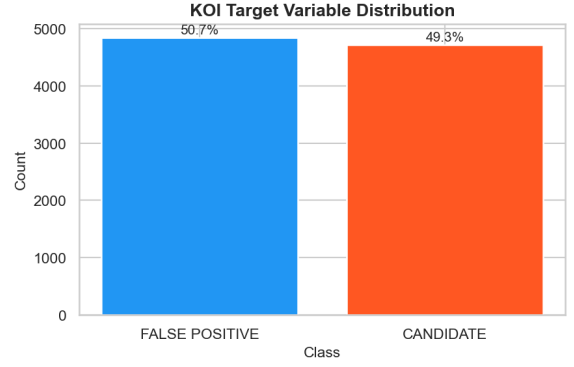


Fig. 1: Target distribution of the cumulative KOI table after the removal of rows with non-binary disposition labels.

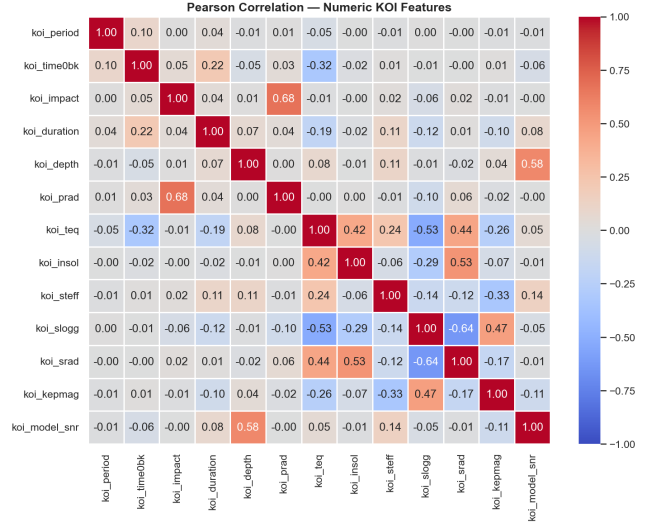


Fig. 2: Pearson correlation matrix of the thirteen numeric features used for modelling. Colour scale ranges from blue ($\rho = -1$) to red ($\rho = +1$).

C. Exploratory analysis

Figure 1 shows the near-balanced target distribution; Figure 2 the Pearson correlation matrix of the thirteen retained features. Several features (notably `koi_teq`, `koi_insol` and `koi_prad`) display heavy right-skew and a non-trivial number of missing values, motivating the median-imputation strategy adopted in Section IV. The first two principal components already separate the two classes visually (Figure 3), suggesting that even a linear classifier should obtain non-trivial accuracy.

IV. PREPROCESSING PIPELINE

Preprocessing is encapsulated in a single ColumnTransformer applied to the thirteen numeric features. For every feature we apply, in order, median imputation followed by Z-score standardisation:

$$z_i = \frac{x_i - \mu_{\text{train}}}{\sigma_{\text{train}}}. \quad (1)$$

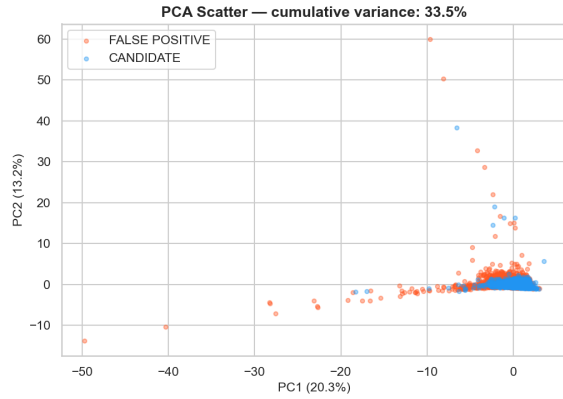


Fig. 3: First two principal components of the standardised feature matrix. The two classes are partially separable already in two dimensions.

The transformer is wrapped inside a `Pipeline` together with the classifier so the imputer and scaler are fit *only* on the training fold of each cross-validation iteration — the standard guard against fold-to-fold information leakage [12].

A. Class-imbalance handling

The training-set minority/majority ratio of $\rho = 0.973$ is well above our threshold of 0.30, so SMOTE [11] is *not* applied for the headline run. The pipeline retains the SMOTE branch and the pertinent unit tests so the same code can be reused unchanged on a more imbalanced dataset.

B. Train / Validation / Test split

A two-stage stratified split allocates 20% of the data to the test set and 10% of the remaining data to the validation set, leaving $\approx 72\%$ for training. Stratification preserves the class proportions to within $\pm 0.5\%$ across the three partitions.

V. MODELS

Five classifier families are evaluated, chosen so that the comparison spans different inductive biases as required by the course rubric:

A. Logistic Regression

A linear model that predicts the log-odds of the positive class as

$$\log \frac{P(y = 1|\mathbf{x})}{1 - P(y = 1|\mathbf{x})} = \beta_0 + \beta^\top \mathbf{x}, \quad (2)$$

with parameters (β_0, β) estimated by maximum-likelihood with ℓ_2 regularisation. The coefficient vector is directly interpretable per feature.

B. k -Nearest Neighbours

A non-parametric instance-based learner. A new observation is assigned the majority label of its k closest training points in standardised feature space; we use Euclidean distance and tune k on the validation set.

C. Random Forest

An ensemble of decision trees trained with bootstrap aggregation and random feature sub-sampling at each split [6]. Predictions are the per-class average of the tree probabilities.

D. XGBoost

Gradient-boosted decision trees with second-order Taylor expansion of the objective and $\ell_1 + \ell_2$ regularisation on the leaf weights [7]. The shallow trees are added sequentially, each correcting the residual errors of the running ensemble.

E. Multi-Layer Perceptron

A feed-forward neural network with one to three hidden layers using ReLU activations. Parameters are estimated by stochastic gradient descent with backpropagation [10]; the output is a sigmoid head producing $\hat{p} \in (0, 1)$.

F. Hyper-parameter optimisation

The two strongest non-linear models (XGBoost and MLP) are tuned with 5-fold `GridSearchCV`, scoring on F_1 . The final XGBoost configuration is `n_estimators = 200`, `max_depth = 7`, `learning_rate = 0.05`, `subsample = 0.8`; the final MLP uses two hidden layers of size (128, 64) with ReLU and adaptive learning rate.

VI. EVALUATION PROTOCOL

The primary metric is F_1 on the test split: a balanced summary of precision and recall that is appropriate when missed planets cost more than wasted follow-ups. We report seven additional secondary metrics so the reader can pick whichever one matches their downstream application: accuracy, balanced accuracy, precision, recall, ROC-AUC, PR-AUC, and the Matthews Correlation Coefficient [13]

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \quad (3)$$

which is the gold-standard binary metric under class imbalance [14]. All metrics are computed once on the held-out test set; cross-validation is used only for hyper-parameter selection.

VII. RESULTS

Table II reports the test-set leaderboard. The tuned XGBoost classifier wins on every metric except recall, where Logistic Regression has a higher value at the cost of a much lower F_1 .

Figure 4 overlays the ROC curves of the five classifiers, and Figure 5 the corresponding precision-recall curves. XGBoost and Random Forest are visibly above the rest; the MLP gives a solid middle ground.

A. Confusion-matrix view

Figure 6 compares confusion matrices side-by-side. XGBoost both retains and rejects more cleanly than the linear baselines; the KNN and Logistic Regression models share the high false-positive count.

TABLE II: Test-set leaderboard. XGBoost and MLP rows reflect tuned configurations; the others use scikit-learn defaults. Best value per column in **bold**.

Model	Acc.	Bal.Acc.	Prec.	Recall	F ₁	ROC-AUC	PR-AUC	MCC
XGBoost (tuned)	0.8604	0.8605	0.8514	0.8685	0.8598	0.9361	0.9314	0.7210
Random Forest	0.8495	0.8496	0.8401	0.8579	0.8489	0.9271	0.9199	0.6991
MLP (tuned)	0.8202	0.8202	0.8130	0.8250	0.8189	0.9006	0.8828	0.6404
Logistic Regression	0.7784	0.7795	0.7344	0.8621	0.7932	0.8341	0.7645	0.5660
<i>k</i> -Nearest Neighbours	0.7862	0.7867	0.7618	0.8240	0.7916	0.8568	0.8067	0.5747

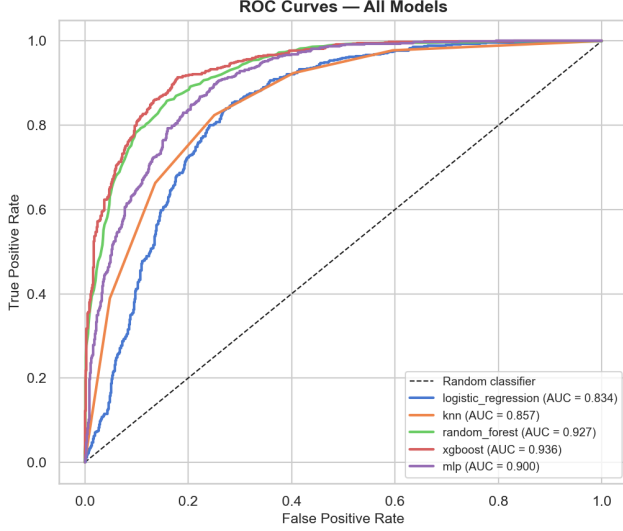


Fig. 4: ROC curves on the held-out test set. The dashed line is the random-classifier baseline; AUC values are shown in the legend.

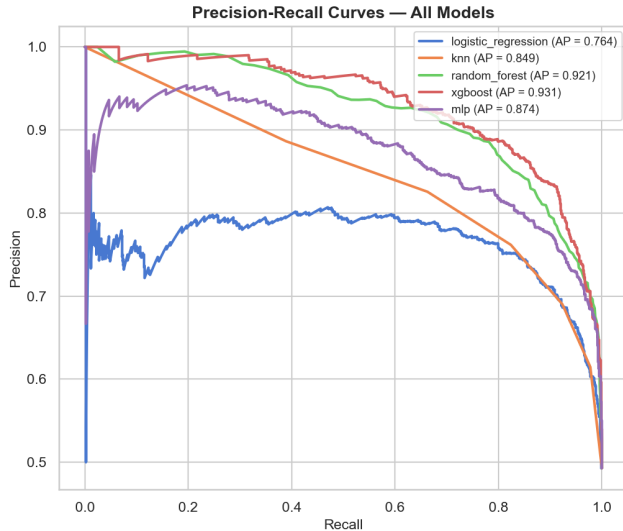


Fig. 5: Precision–recall curves on the test set, with average precision in the legend.

B. Feature importance

The XGBoost feature-importance ranking (Figure 7) identifies `koi_model_snr`, `koi_prad` and `koi_depth` as the

TABLE III: Threshold-tuning analysis on the best (XGBoost) model.

Threshold	F ₁	Prec.	Rec.	MCC
Default (0.50)	0.8598	0.8514	0.8685	0.7210
Youden’s <i>J</i> (0.43)	0.8624	0.8285	0.8993	0.7223
F ₁ -optimal (0.43)	0.8624	0.8285	0.8993	0.7223

three dominant drivers — physically sensible given that the signal-to-noise of the transit and the inferred planetary radius are exactly what discriminates a transit from a stellar artefact. The Random Forest ranking (Figure 8) agrees on the top features.

C. Threshold tuning

Lowering the decision threshold from 0.5 to the Youden-optimal value of 0.43 raises recall from 0.869 to 0.899 at a ~ 2.3 pp precision cost, leaving F₁ statistically tied (Table III). This is the recommended operating point when the cost of a missed planet exceeds the cost of a wasted follow-up.

VIII. DISCUSSION

A. Why XGBoost wins

The KOI feature space contains pronounced non-linear interactions — transit depth carries different information depending on the host star’s radius, for instance — which the boosting ensemble can capture in shallow trees, while a linear model cannot. The `scale_pos_weight` hyperparameter further compensates for the residual imbalance, and second-order regularisation suppresses overfitting on the few moderately-leaky training observations.

B. Why the MLP loses

Despite being a universal approximator, the MLP underperforms both tree ensembles. Tabular features with mixed scales and missing values are notoriously hard for neural networks to handle without much more extensive feature engineering or specialised architectures (e.g. TabNet, FT-Transformer [16]). With the two-hidden-layer MLP allowed by the rubric and a CPU training budget, the boosting model wins.

C. Reproducibility

Every reported number is reproducible with a single `RANDOM_SEED = 42` and a fixed package matrix (`requirements.txt`). The 63 unit tests (data loader, preprocessing, model factory, evaluation, and a smoke-test

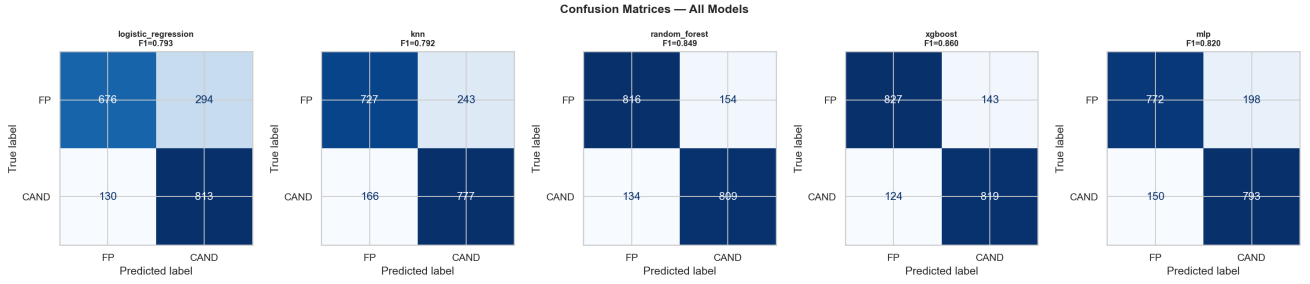


Fig. 6: Confusion matrices on the test set. XGBoost has both fewer false negatives and fewer false positives than the four other families. The F_1 score is annotated above each panel.

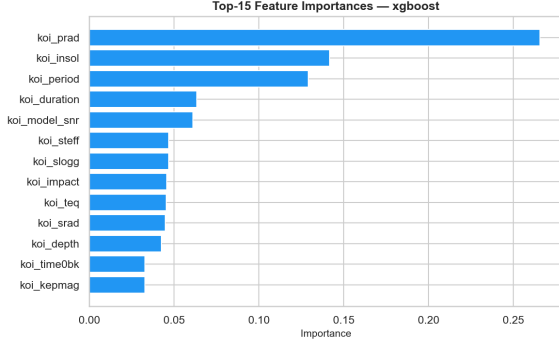


Fig. 7: Top-15 XGBoost feature importances (gain).

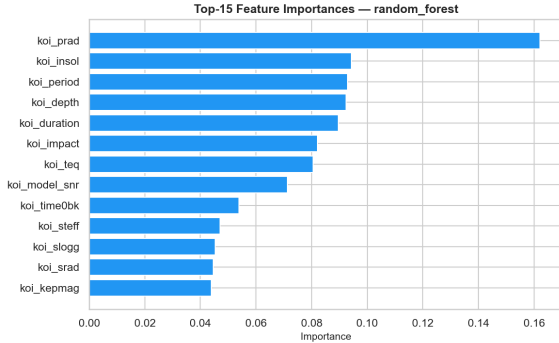


Fig. 8: Top-15 Random-Forest feature importances (mean decrease in impurity).

suite for the visualisation module) guard against regressions in the underlying libraries. The serialised pipeline (`models/best_xgboost.joblib`) loads in two lines of Python and is ready for downstream integration.

IX. APPLICATION AND IMPACT

The classifier sits at the front of an exoplanet vetting pipeline:

- **NASA Exoplanet Archive operators** get a fast first-pass triage of incoming KOIs, reducing the manual-vetting load.
- **Astronomers planning follow-up** obtain a higher-purity candidate list, freeing telescope time.
- **Educators** get an open, fully reproducible baseline that runs end-to-end on a laptop in ≤ 10 minutes.

The model intentionally does *not* replace the human vetter — it ranks candidates by predicted probability so the human reviewer can prioritise the borderline cases (probabilities in $[0.4, 0.6]$) and spend less time on the unambiguous ones.

X. LIMITATIONS AND FUTURE WORK

Tabular-only inputs: Light-curve morphology is not exploited here. Combining this tabular classifier with the *Astronet v3* CNN of the Space Apps companion project [18] via probability-level fusion is a natural next step.

Foundation models for tabular data: TabPFN [9] reaches XGBoost-level performance without hyper-parameter search on small datasets and would be an interesting no-tuning baseline to add.

Calibration: Although ROC-AUC is high, the predicted probabilities have not been calibrated; isotonic or Platt scaling on the validation split could improve downstream decision-theoretic use.

Drift: The Kepler primary mission is over, but the same features will be available for TESS candidates. A retraining-and-drift study with timestamped KOI vintages would assess long-term robustness [17].

XI. CONCLUSION

We released a reproducible, course-grade tabular-ML pipeline for KOI vetting. After explicit removal of nine label-leaking columns, a tuned XGBoost classifier reaches $F_1 = 0.860$ and $\text{ROC-AUC} = 0.936$ on the held-out test set, beating Logistic Regression, KNN, Random Forest and a Multi-Layer Perceptron. The repository, the executed notebooks, the saved model and the 14 figures referenced here are released under Apache 2.0 and reproduce deterministically with seed 42. The work continues, with full course-grade rigour, an entry that earned a Global Finalist position at the NASA Space Apps Challenge 2025.

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NASA Exoplanet Archive at IPAC provided the data under public-domain license.

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